

NLSS IV poverty calculations and replication guide

Data preparation, weighted estimators, Gini and Lorenz construction, validation checks, exact results, and the complete Stata program.

Version 1.0 | 21 June 2026 | Software: Stata 18 MP

Reproducibility note. The task refers to `weight.dta`; the supplied workspace instead contains `poverty.dta`. Its labels identify calibrated household and individual weights, poverty status, consumption, poverty line and analytical domains, so it is treated as the intended file. The source files are never modified.

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1. Inputs and analytical population

File	Role	Key variables
poverty.dta	9,600-household poverty and weight file	psu_number hh_number domain ad_4 hsize hhs_wt ind_wt pcep pline poor
S01.dta	46,870 roster records; used only to identify the unique head and head sex	psu_number hh_number q01_02 q01_04

The household is the stored observation in the analysis file. The programme retains the roster record with relationship-to-head code `q01_04==1`, verifies one head per household, and merges it one-to-one on PSU and household number. All 9,600 households match.

Core definitions

- **Welfare aggregate:** spatially deflated annual per-capita consumption expenditure (*pcep*).
- **Poverty line:** NPR 72,908.05 per person per year (*pline*).
- **Poverty status:** $h_i = 1$ when $pcep_i < pline$; the programme asserts this equals supplied *poor*.
- **Household weight:** $w_i = hhs_wt$, calibrated to the 2021 final household count.
- **Person weight:** $v_i = w_i n_i = ind_wt$, where n_i is household size.
- **Urban:** Kathmandu Valley Urban plus Other Urban (*ad_4* 1 or 2); Rural is code 3.

2. Methodology and equations

Weighted poverty headcount

$$\text{Households: } H^{HH} = \sum_i w_i h_i / \sum_i w_i$$

$$\text{Persons: } H^P = \sum_i v_i h_i / \sum_i v_i, \quad v_i = w_i n_i$$

The household rate answers ‘‘what share of households is poor?’’ The person rate answers ‘‘what share of people lives in poor households?’’ The latter is the conventional population poverty headcount.

Lorenz curve and weighted Gini

Order households by ascending *pcep*. Using person weights, define cumulative population share X_i and cumulative consumption share L_i :

$$X_i = (\sum_{j \leq i} v_j) / (\sum_j v_j) \quad \text{and} \quad L_i = (\sum_{j \leq i} v_j y_j) / (\sum_j v_j y_j)$$

where y_i is real per-capita consumption. The do-file integrates the Lorenz curve by trapezoids:

$$G = 1 - \sum_i (L_i + L_{i-1})(X_i - X_{i-1}), \text{ with } X_0 = L_0 = 0.$$

All reported Ginis describe inequality among persons in the relevant group, with each household's per-capita consumption repeated conceptually for its weighted population.

3. Calculation workflow

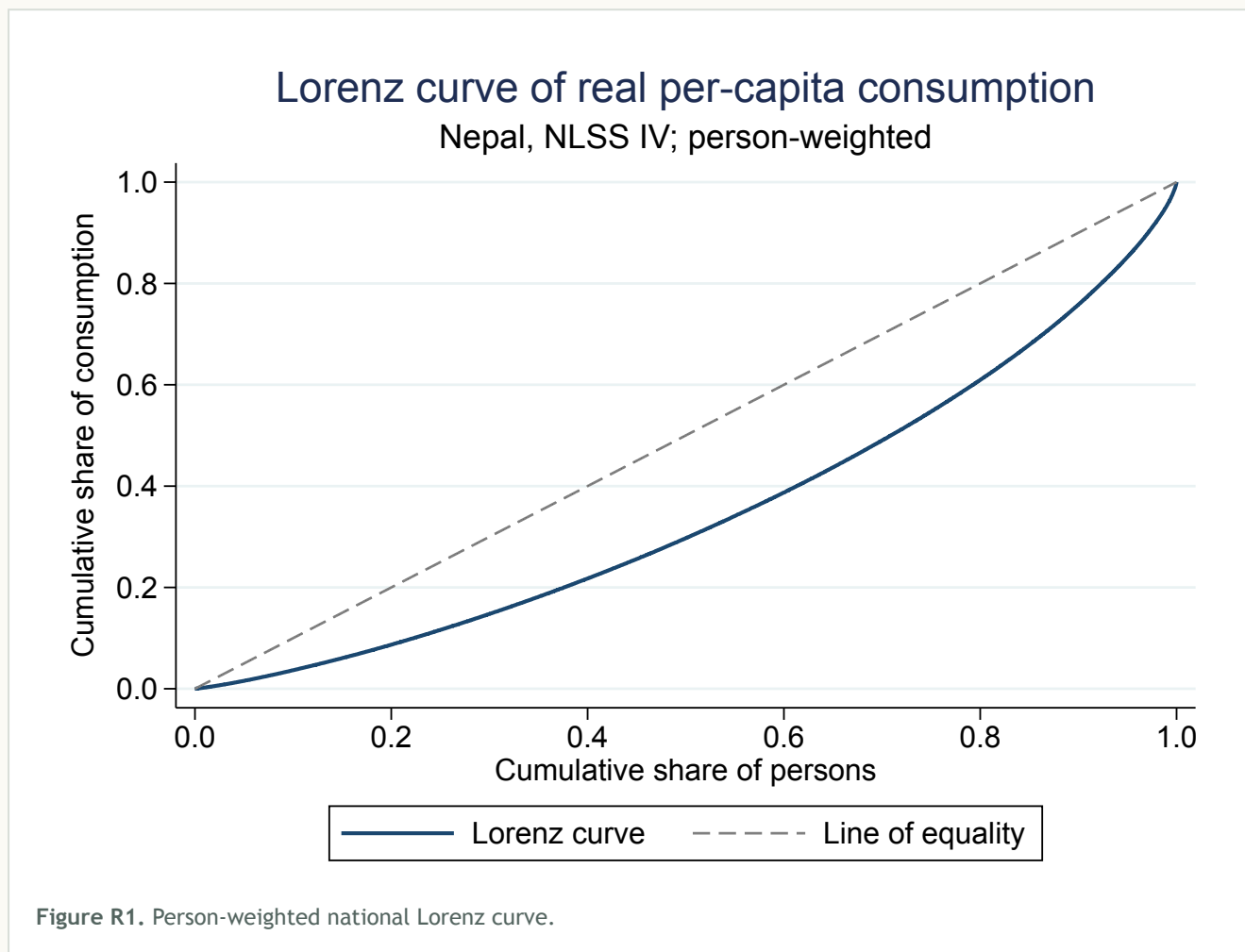
1. Extract unique household heads from `S01.dta`.
2. Merge head sex into `poverty.dta` and assert a complete match.
3. Verify poverty status and the person-weight identity.
4. Create the two-category urban/rural variable.
5. Compute household and person headcounts plus the person-weighted Gini nationally and by residence, head gender, and 15 analytical domains.
6. Export exact results to CSV/DTA and graph the Lorenz curve and subgroup headcounts in Stata.
7. Build the self-contained HTML outputs from those machine-readable results.

4. Exact and publication-rounded results

Area/group	Weighted households	Weighted persons	Household poverty (%)	Individual poverty (%)	Gini
Nepal	7,185,103	28,740,504	16.7	20.3	0.300
Urban	5,023,805	19,966,576	15.2	18.3	0.303
Rural	2,161,298	8,773,927	20.2	24.7	0.287
Male	4,518,910	19,388,104	16.9	20.3	0.303
Female	2,666,193	9,352,400	16.3	20.3	0.294
Koshi Urban	804,093	3,067,096	13.4	15.9	0.284
Koshi Rural	412,798	1,608,214	16.6	19.7	0.313
Madhesh Urban	1,010,251	4,838,320	18.9	21.7	0.302
Madhesh Rural	337,626	1,647,827	21.6	25.0	0.242
Kathmandu Valley Urban	845,255	3,091,107	6.1	7.4	0.266
Bagmati Urban(Excluding KTM Valley)	486,012	1,751,007	12.8	14.1	0.314
Bagmati Rural	291,642	1,027,367	20.1	25.6	0.327
Gandaki Urban	473,651	1,640,114	10.9	12.6	0.280
Gandaki Rural	221,412	756,614	7.9	10.3	0.261
Lumbini Urban	817,117	3,232,770	19.1	24.1	0.322
Lumbini Rural	537,286	2,214,608	20.3	24.7	0.270
Karnali Urban	209,921	797,408	18.4	23.2	0.330
Karnali Rural	162,875	675,707	25.5	30.9	0.266

Area/group	Weighted households	Weighted persons	Household poverty (%)	Individual poverty (%)	Gini
Sudurpaschim Urban	377,506	1,548,754	27.2	30.9	0.291
Sudurpaschim Rural	197,659	843,590	35.0	40.2	0.245

Percent columns are rounded to one decimal and Ginis to three decimals for display. The CSV retains additional precision. Weighted counts are rounded to whole units.



5. Quality assurance and limitations

- **PASS** 9,600 unique household keys in the poverty file.
- **PASS** Exactly 9,600 unique heads in the roster.
- **PASS** Complete one-to-one merge; zero unmatched records.
- **PASS** `poor == (pcep < pline)` for every household.
- **PASS** `ind_wt == hhs_wt * hhsiz` within floating-point tolerance.
- **PASS** Weighted totals: 7,185,103 households and 28,740,504 persons.

Limitations. The supplied data provide a single cross-section, so the analysis documents spatial and demographic differences, not change over time or causal effects. Stratum identifiers were not supplied; therefore this brief reports weighted point estimates, not design-based standard errors or confidence intervals. Head-gender comparisons classify every household member by the sex of the head and are not estimates of poverty by each person's sex. Domain estimates have smaller samples and should be interpreted with greater caution.

6. Rerun instructions

1. Place `poverty.dta`, `S01.dta`, `nlss4_poverty_analysis.do`, and `build_reports.ps1` in the same folder.
2. Set Stata's working directory to that folder.
3. Run `do nlss4_poverty_analysis.do`.
4. From PowerShell, run `powershell -ExecutionPolicy Bypass -File .\build_reports.ps1`.
5. Confirm that the Stata log ends with "Analysis complete" and review files under `output/results`, `output/figures`, and `output`.

Expected outputs

Output	Purpose
<code>poverty_results.csv / .dta</code>	Exact estimates for all requested groups
<code>lorenz_coordinates.csv</code>	National Lorenz coordinates
Three SVG files	Stata-generated figures
<code>policy_brief.html</code>	Self-contained brief for the Prime Minister
<code>replication_guide.html</code>	This self-contained internal guide

7. Complete Stata code

The following is the exact master do-file used for the estimates and graphs.

```
version 18.0
clear all
set more off
set linesize 120
set scheme s2color

capture mkdir "output"
capture mkdir "output/figures"
```

```

capture mkdir "output/results"

/*
  NLSS IV poverty dynamics: complete replication program
  Inputs: poverty.dta (the supplied weight/poverty file) and S01.dta
  Unit of analysis: household. Person estimates use hhs_wt * hhsz.
*/

* Extract the sex of the unique household head.
use "S01.dta", clear
keep if q01_04 == 1
isid psu_number hh_number
keep psu_number hh_number q01_02
rename q01_02 head_gender
label variable head_gender "Sex of household head"
tempfile heads
save `heads'

* Merge head characteristics into the poverty/weight file.
use "poverty.dta", clear
isid psu_number hh_number
merge 1:1 psu_number hh_number using `heads', assert(match) nogen
assert poor == (pcep < pline)
assert abs(ind_wt - hhs_wt * hhsz) < .1

generate byte urban = inlist(ad_4, 1, 2)
label define urban_lab 0 "Rural" 1 "Urban"
label values urban urban_lab
label variable urban "Urban/rural residence"

tempfile analysis
save `analysis'

* Weighted Gini from the trapezoidal area under the Lorenz curve.
capture program drop weighted_gini
program define weighted_gini, rclass
    syntax varname, Wvar(varname)
    preserve
        keep `varlist' `wvar'
        drop if missing(`varlist', `wvar') | `wvar' <= 0
        sort `varlist'
        quietly generate double __wx = `wvar' * `varlist'
        quietly summarize `wvar', meanonly
        local W = r(sum)
        quietly summarize __wx, meanonly
        local Y = r(sum)
        quietly generate double __L = sum(__wx) / `Y'
        quietly generate double __L0 = __L[_n-1]
        quietly replace __L0 = 0 in 1
        quietly generate double __area2 = (__L + __L0) * (`wvar' / `W')
        quietly summarize __area2, meanonly
        return scalar gini = 1 - r(sum)
        return scalar N = _N
    restore
end

tempname results_post
tempfile results
postfile `results_post' str24 section str48 group int display_order ///

```

```

    long sample_households double households persons hc_household hc_individual gini
    ///
    using `results', replace
    global NLSS_RESULTS_POST `results_post'

    capture program drop post_estimate
    program define post_estimate
        syntax, Section(string) Group(string) Order(integer) Condition(string)
        preserve
            quietly keep if `condition'
            quietly count
            local n = r(N)
            quietly summarize hhs_wt, meanonly
            local households = r(sum)
            quietly summarize ind_wt, meanonly
            local persons = r(sum)
            quietly summarize poor [aw=hhs_wt], meanonly
            local hc_hh = r(mean)
            quietly summarize poor [aw=ind_wt], meanonly
            local hc_ind = r(mean)
            quietly weighted_gini pcep, wvar(ind_wt)
            local gini = r(gini)
            post $NLSS_RESULTS_POST ("`section'") ("`group'") (`order') (`n') ///
                (`households') (`persons') (`hc_hh') (`hc_ind') (`gini')
        restore
    end

    use `analysis', clear
    post_estimate, section("National") group("Nepal") order(1) condition("1")

    foreach u in 1 0 {
        local ulab : label urban_lab `u'
        local ord = cond(`u' == 1, 1, 2)
        post_estimate, section("Urban/rural") group("`ulab'") order(`ord')
        condition("urban == `u'")
    }

    foreach g in 1 2 {
        local glab : label Q01_02_VS1 `g'
        post_estimate, section("Head gender") group("`glab'") order(`g')
        condition("head_gender == `g'")
    }

    levelsof domain, local(domains)
    local ord = 0
    foreach d of local(domains) {
        local ++ord
        local dlab : label domainlab `d'
        post_estimate, section("Analytical domain") group("`dlab'") order(`ord')
        condition("domain == `d'")
    }
    postclose `results_post'
    macro drop NLSS_RESULTS_POST

    use `results', clear
    sort section display_order
    format households persons %15.0fc
    format hc_household hc_individual gini %9.6f
    export delimited using "output/results/poverty_results.csv", replace

```

```

save "output/results/poverty_results.dta", replace

* Chart 1: National person-weighted Lorenz curve.
use `analysis', clear
sort pcep
generate double __wx = ind_wt * pcep
quietly summarize ind_wt, meanonly
generate double population_share = sum(ind_wt) / r(sum)
quietly summarize __wx, meanonly
generate double consumption_share = sum(__wx) / r(sum)
local newN = _N + 1
set obs `newN'
replace population_share = 0 in `newN'
replace consumption_share = 0 in `newN'
sort population_share
twoway (line consumption_share population_share, sort lcolor(navy) lwidth(medthick))
///
      (function y=x, range(0 1) lcolor(gs8) lpattern(dash)), ///
      title("Lorenz curve of real per-capita consumption") ///
      subtitle("Nepal, NLSS IV; person-weighted") ///
      xtitle("Cumulative share of persons") ytitle("Cumulative share of consumption")
///
      xlabel(0(.2)1, format(%3.1f)) ylabel(0(.2)1, format(%3.1f) angle(horizontal))
///
      legend(order(1 "Lorenz curve" 2 "Line of equality") rows(1)) ///
      graphregion(color(white)) plotregion(color(white)) name(lorenz, replace)
graph export "output/figures/lorenz_curve.svg", replace
export delimited population_share consumption_share using ///
      "output/results/lorenz_coordinates.csv", replace

* Chart 2: Individual poverty headcount by analytical domain.
use "output/results/poverty_results.dta", clear
keep if section == "Analytical domain"
generate double hc_percent = 100 * hc_individual
graph hbar (asis) hc_percent, over(group, sort(1) descending label(labsize(small)))
///
      title("Poverty prevalence varies sharply across Nepal") ///
      subtitle("Individual headcount ratio by analytical domain (%)") ///
      ytitle("Percent of persons below the poverty line") ///
      ylabel(0(5)35, angle(horizontal)) blabel(bar, format(%4.1f) size(vsmall)) ///
      bar(1, color(navy)) graphregion(color(white)) plotregion(color(white)) ///
      name(domain_poverty, replace)
graph export "output/figures/domain_poverty.svg", replace

* Chart 3: Poverty and inequality by urban/rural residence.
use "output/results/poverty_results.dta", clear
keep if section == "Urban/rural"
generate double hc_percent = 100 * hc_individual
graph bar (asis) hc_percent, over(group) ///
      title("Rural and urban poverty prevalence") ///
      subtitle("Individual headcount ratio (%)") ytitle("Percent") ///
      ylabel(0(5)30, angle(horizontal)) blabel(bar, format(%4.1f)) ///
      bar(1, color(navy)) graphregion(color(white)) plotregion(color(white)) ///
      name(urban_poverty, replace)
graph export "output/figures/urban_rural_poverty.svg", replace

display as result "Analysis complete. Results are in output/results and figures in

```



```
output/figures."  
exit, clear
```